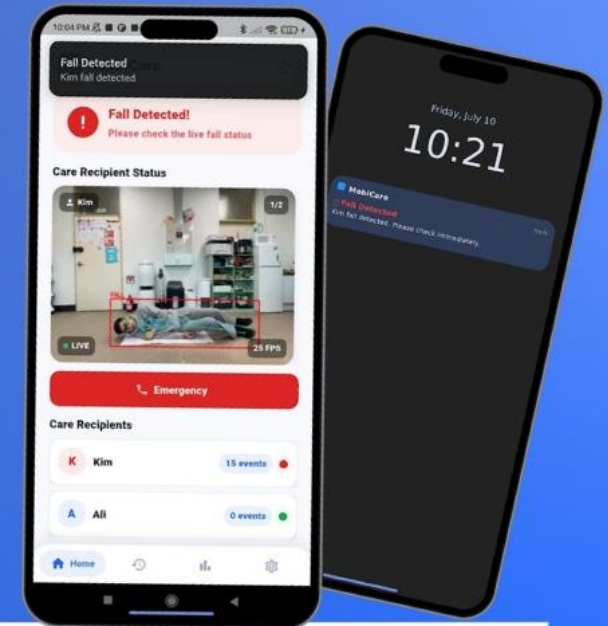




# Edge AI-Based Real-Time Fall Detection and Alert System for Elderly People Living Alone

24/7 Real-Time Monitoring and Instant Response Platform for Elderly Living Alone in a Super-Aged Society



Team - Choco Haim

Damin Kim, Jonghyeok Park, Aliboev Abbos, Seongwoo Cho

# Team



**Aliboev Abbos**

AI Development



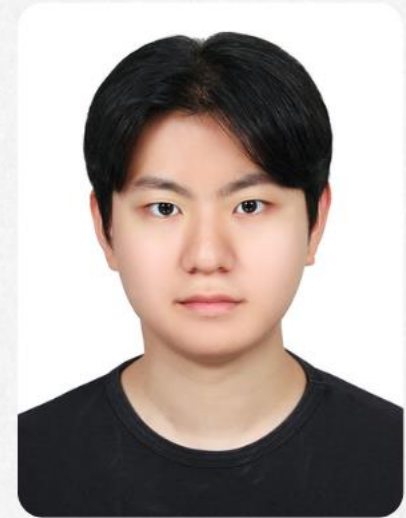
**Damin Kim**

Mobile Development



**Jonghyeok Park**

Kinematics Algorithm



**Seongwoo Cho**

AI Development



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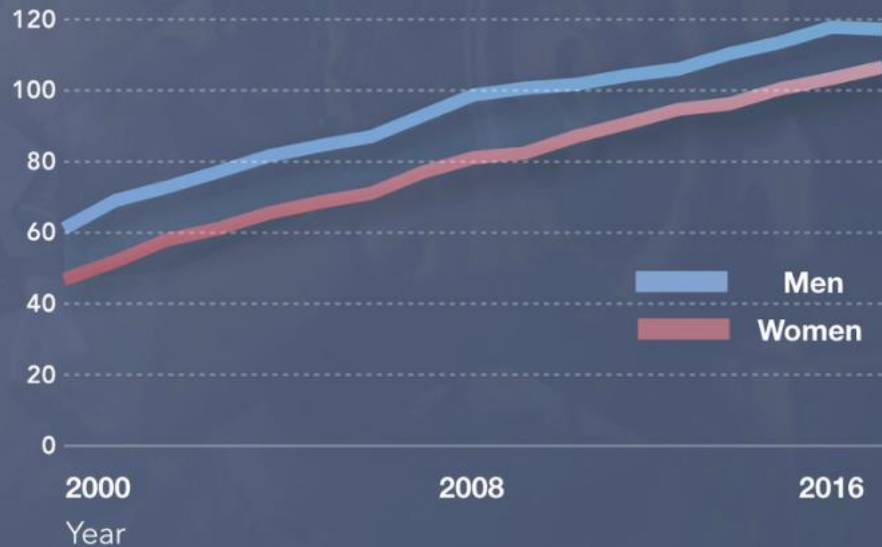
- Accuracy / F1 improvement
- Personal, social, technical impact
- Conclusion and Future Work

# Research Background and Motivation

## Mortality Rate from Falls

Falls are a leading cause of death for the elderly. More elderly men and women are dying from falls every year.

Mortality Rate (per 100,000)



Clearvue Health

Harholdt et al

## Fall Incidence Among Adults 65+

- 20% of adults aged 65+ **experience falls**
- Fall-related **deaths** are increasing
- Many older adults **live alone**
- Fast detection **saves lives**





# Research Background and Motivation

**56%**

**Where?**

**Living Room, bedroom**



**at home**

**64%**



**Public roads**

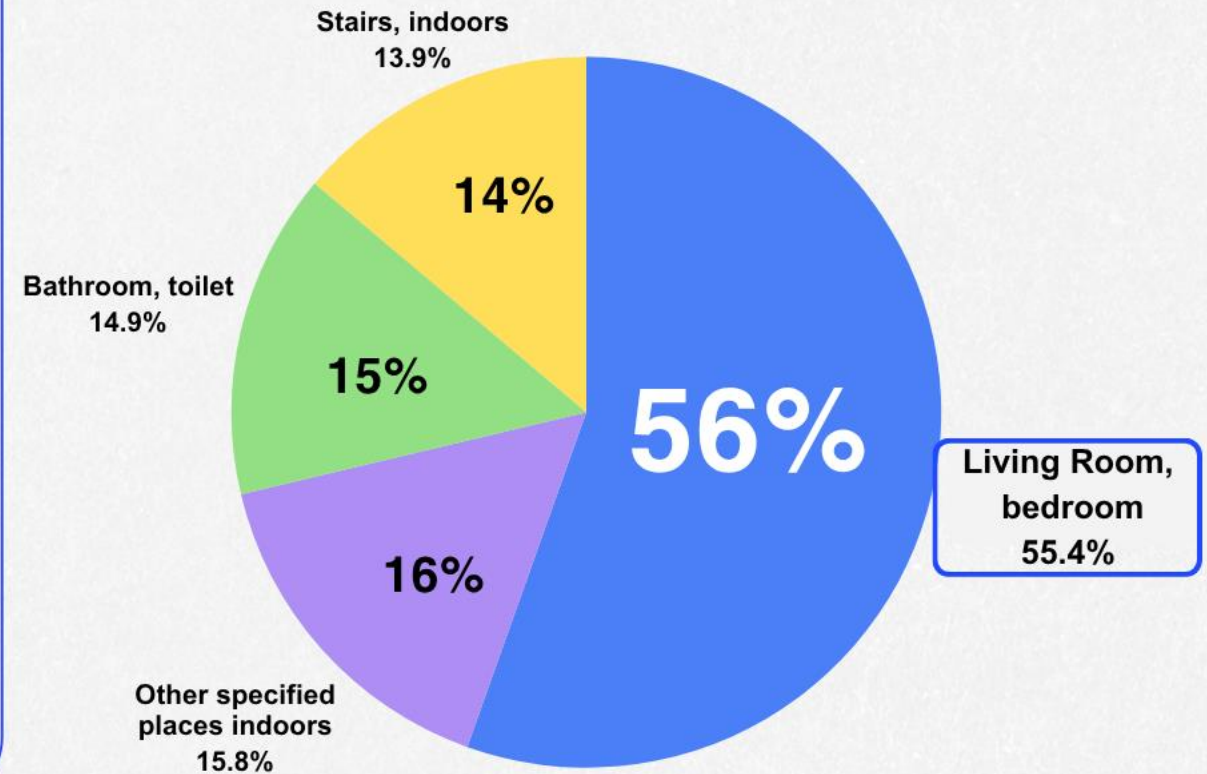
**16%**



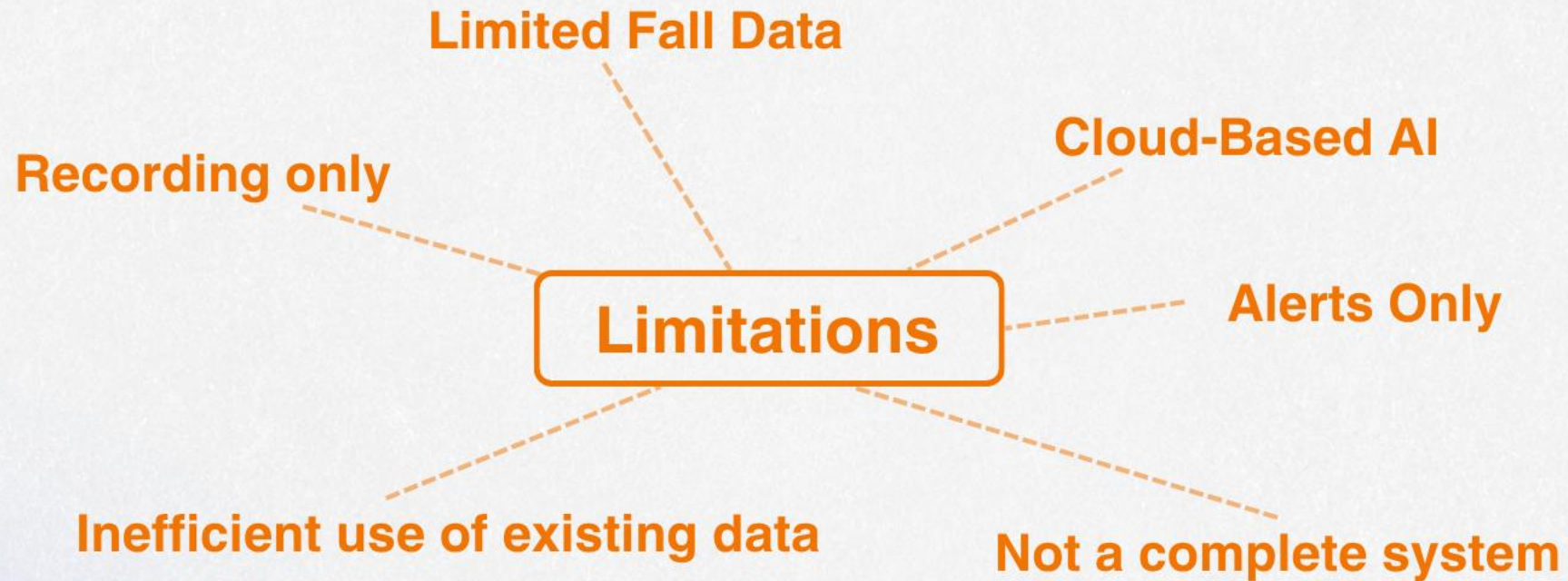
**Residential  
institution**

**9 %**

## In-Home Fall locations

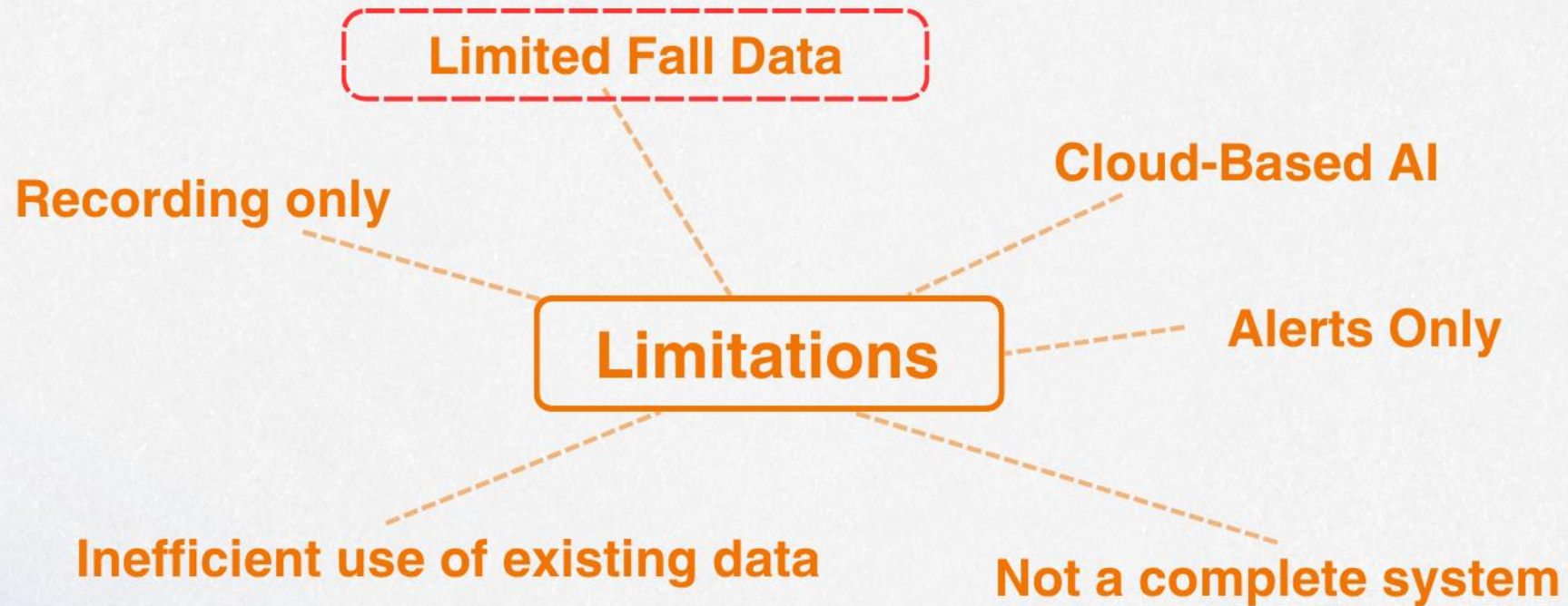


# Limitations of Existing Solutions

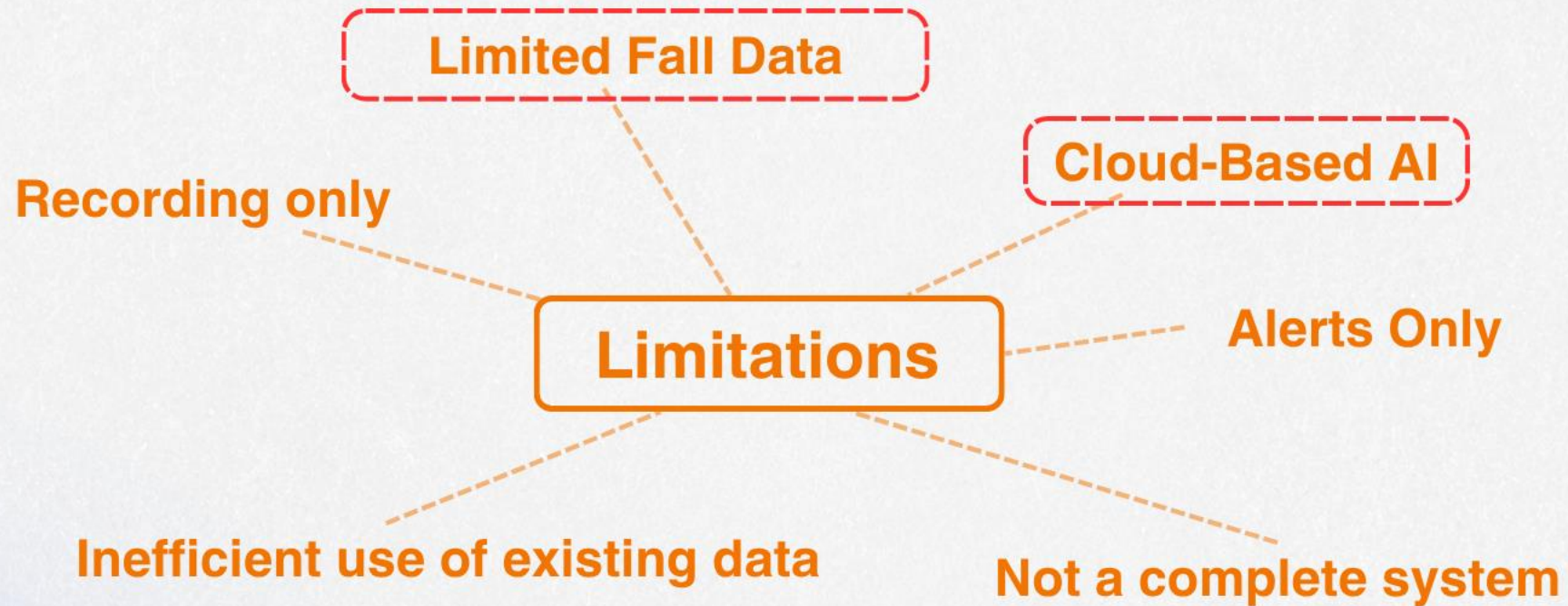




# Limitations of Existing Solutions

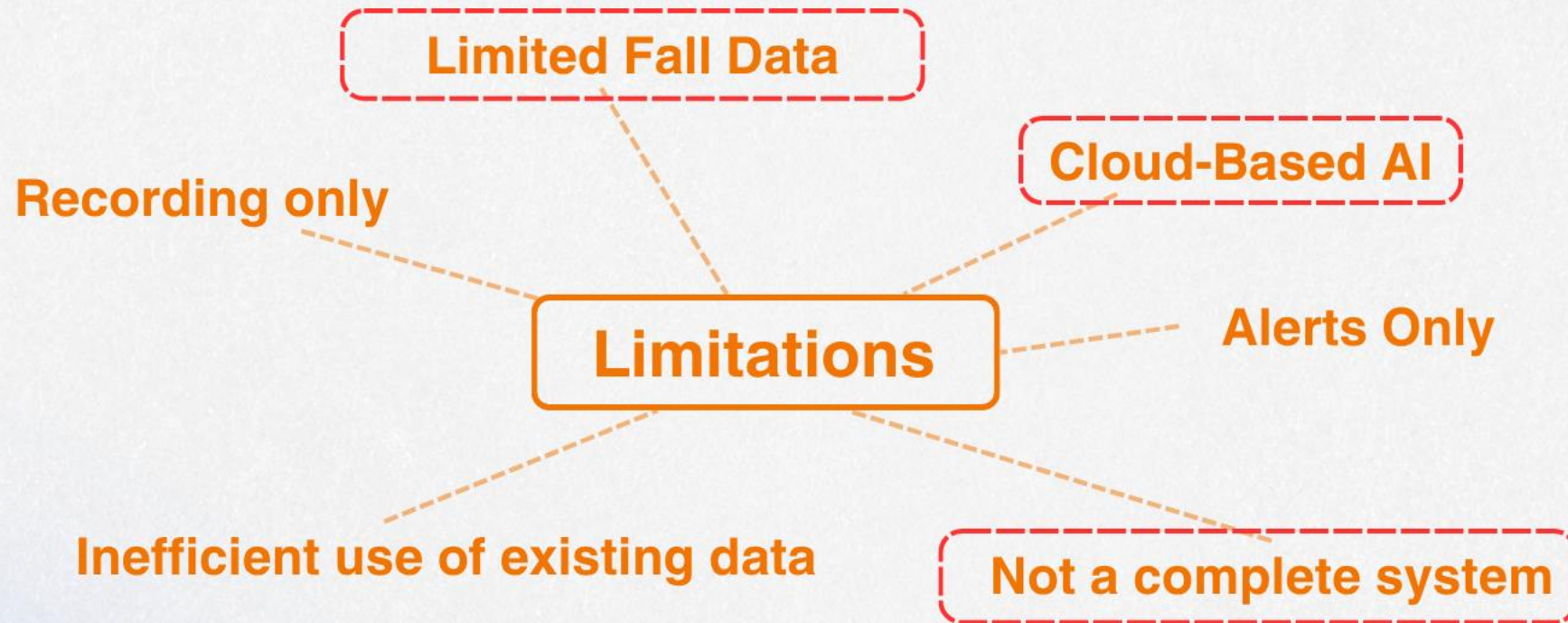


# Limitations of Existing Solutions

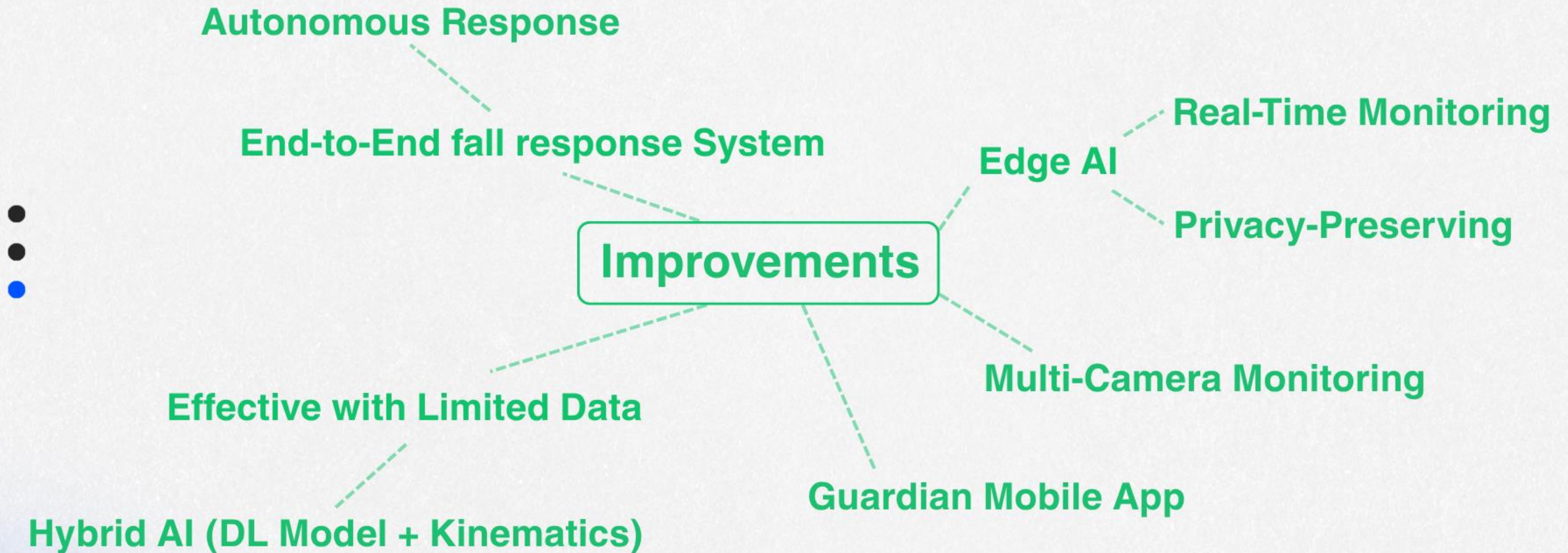




# Limitations of Existing Solutions

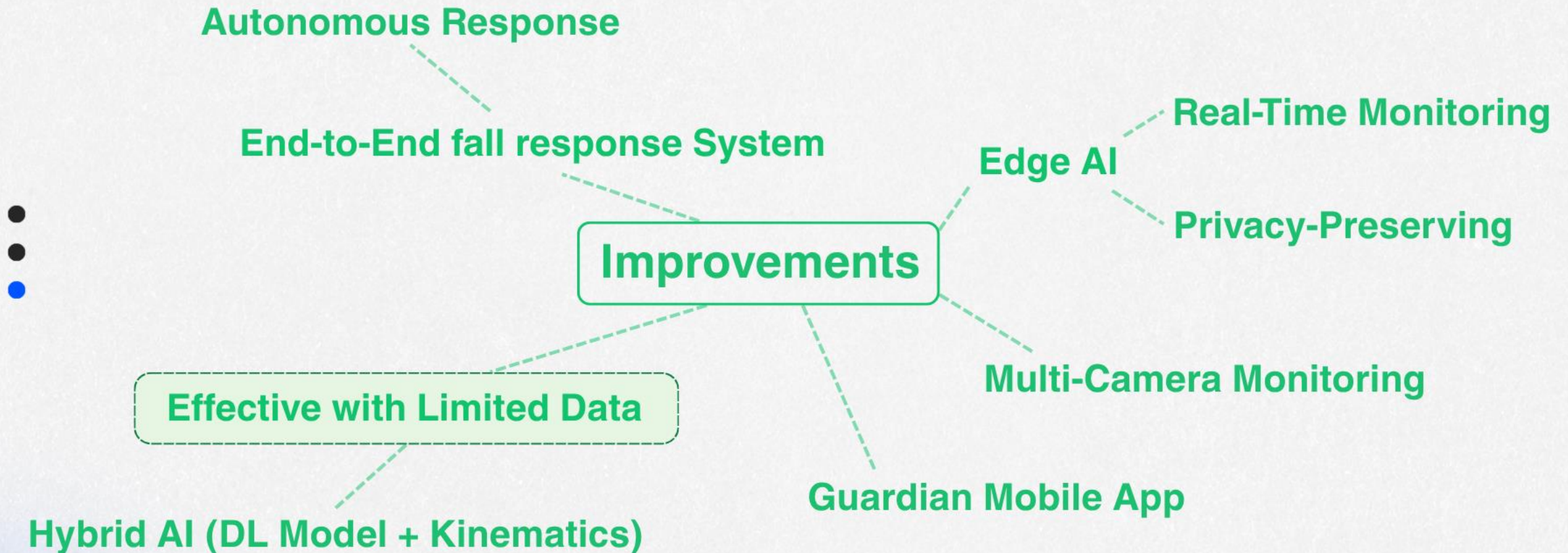


# Our Improvements

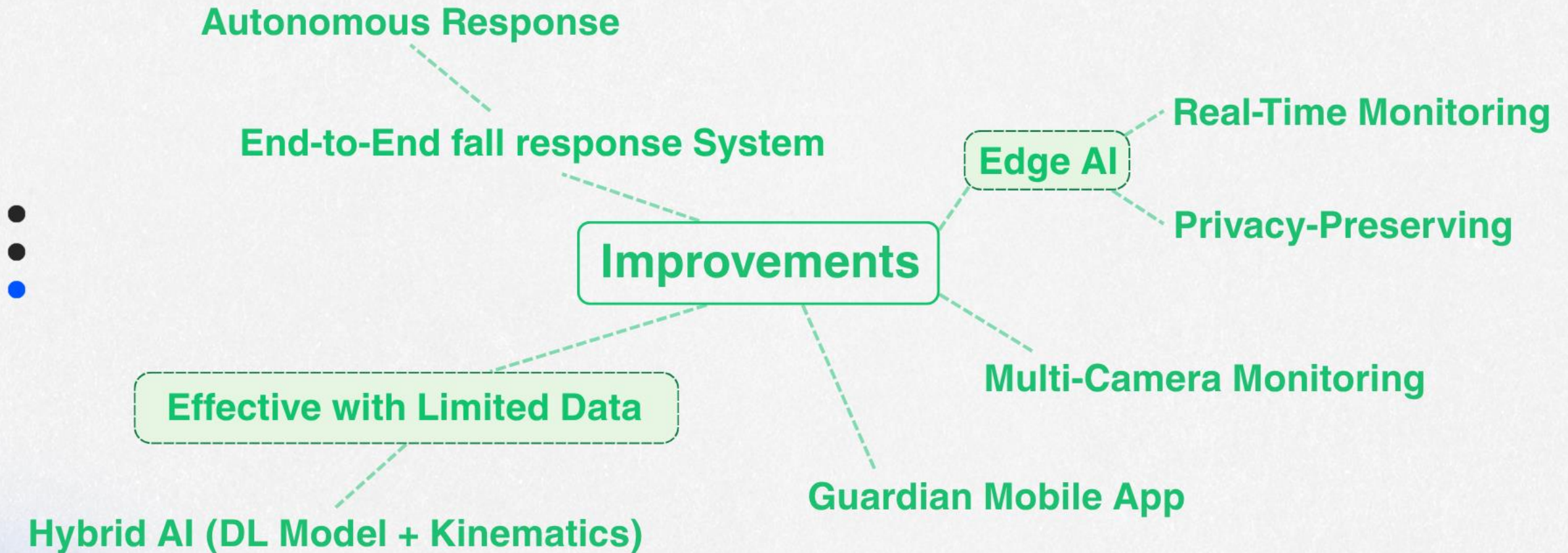




# Our Improvements

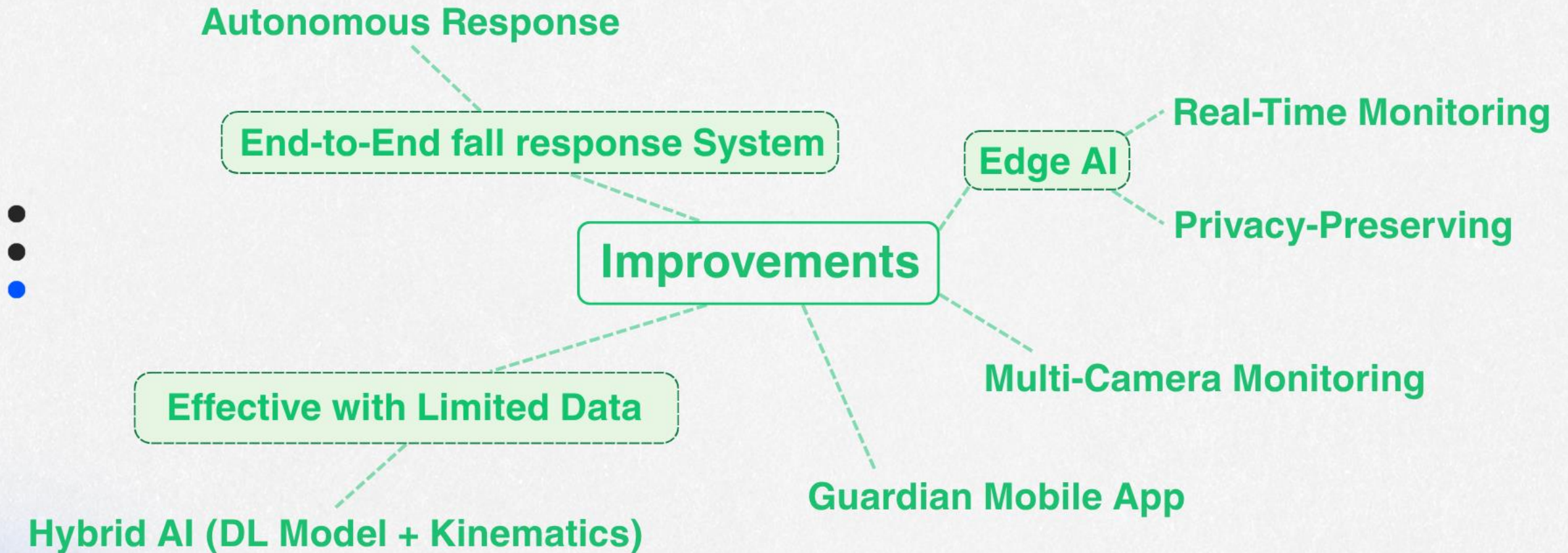


# Our Improvements

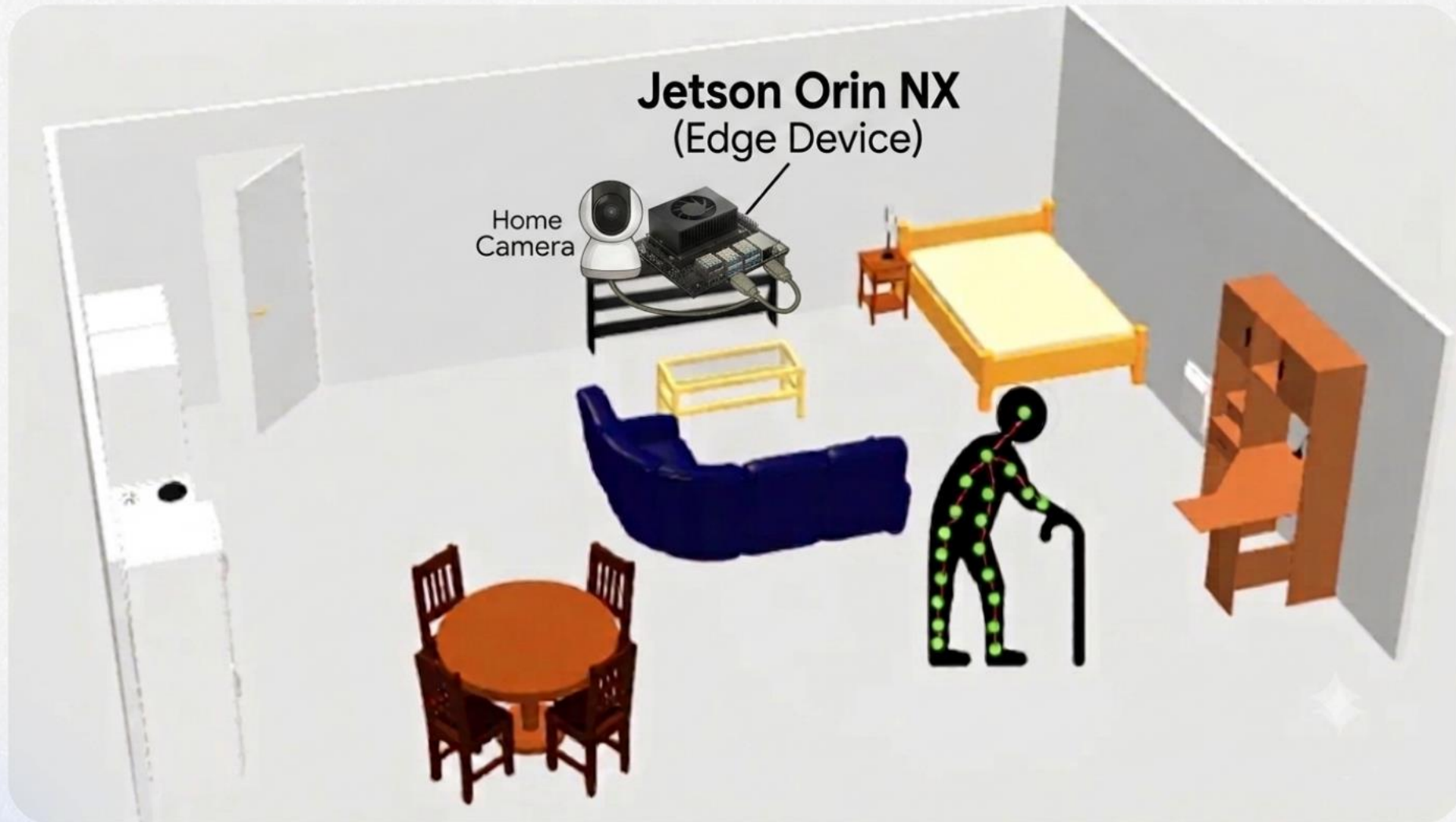




# Our Improvements



# System Overview

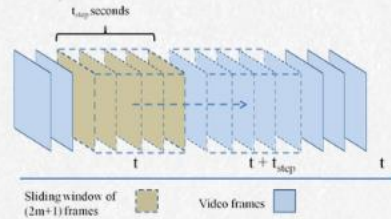




# Main Process



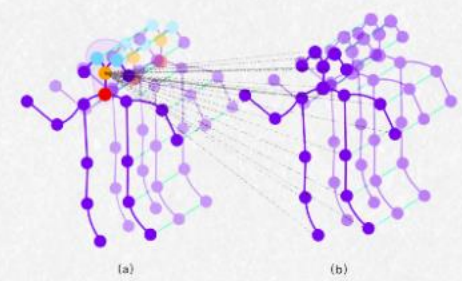
Video Frame



Sliding Window



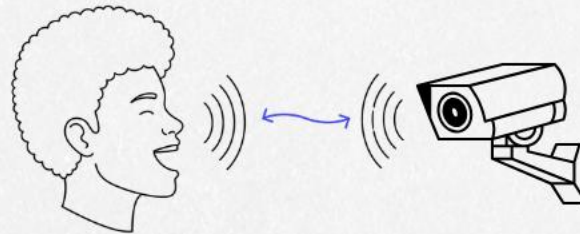
YOLO11n-pose



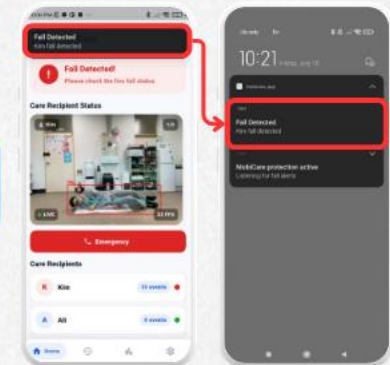
ST-GCN  
(step-1)



Kinematic Filter  
(step-2)

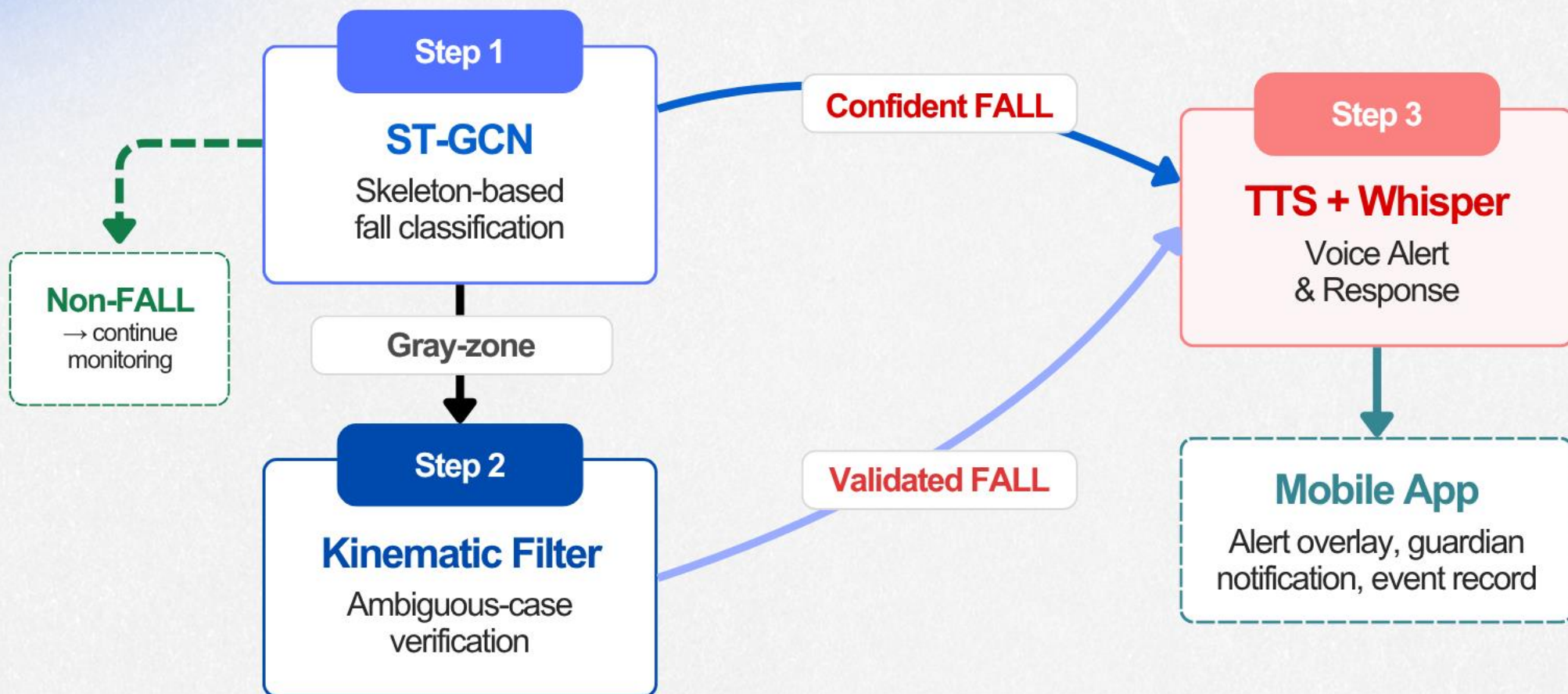


TTS & STT  
(step-3)



Fall Alert  
(SMS & Mobile App)

# Core Decision Flow

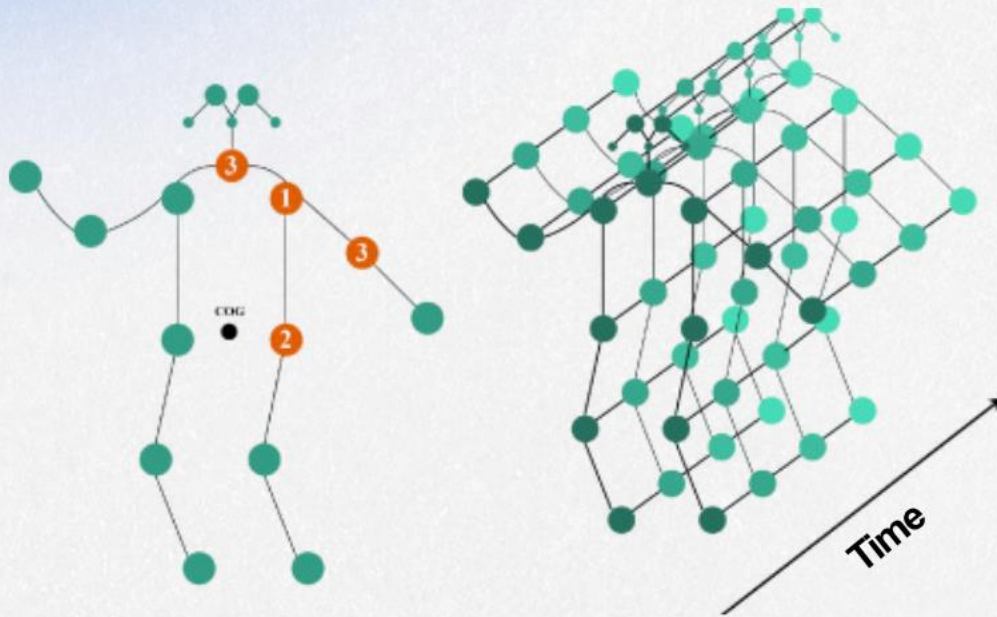


Multi-step validation completely fixes the false alarm issue, triggering SMS and app alarm only during real emergencies.



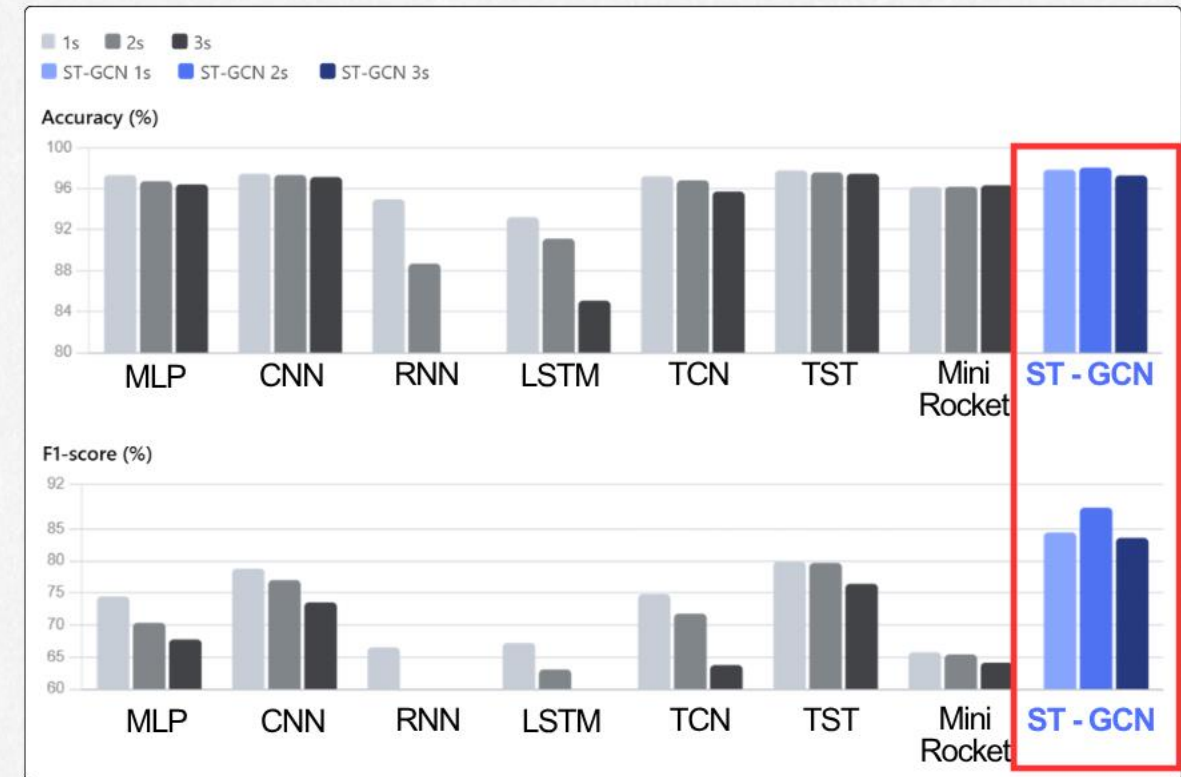
# Why ST-GCN ?

Step - 1



- Tracks body movement over time
- Can see how fall happens step by step
- Ignores background information
- Lightweight, runs on small devices

**Figure** Performance of eight different fall-detection algorithms using different window sizes: 1 s, 2 s, and 3 s.

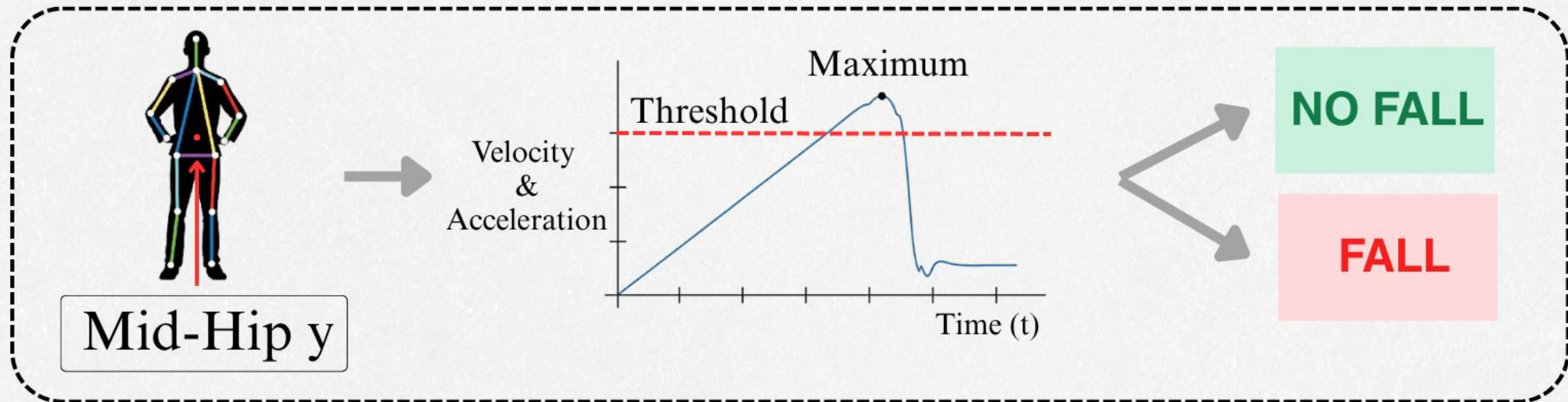
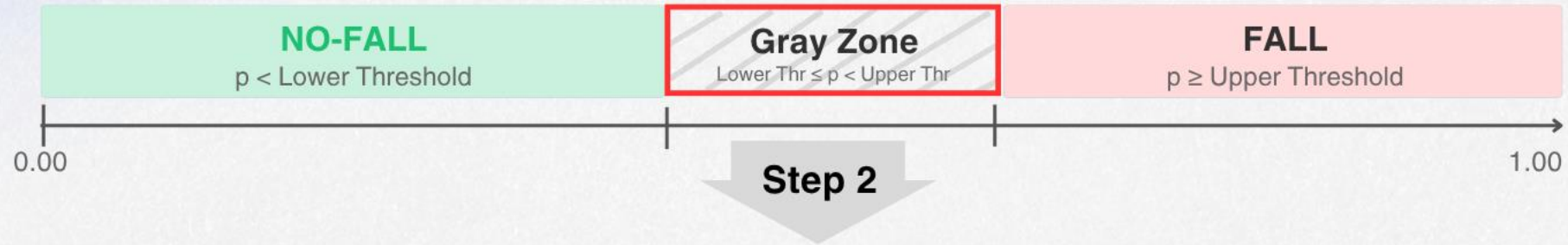


**ST-GCN** achieved the highest and **most stable accuracy** in Fall Detection case

# Kinematic Filter

Step - 2

$p$  is Probability calculated by “ST-GCN in Step 1”



- ST-GCN outputs within the **Gray Zone trigger** velocity and acceleration evaluation
- **Mid-hip velocity and acceleration** exceeding the threshold are classified as a Fall, otherwise Not Fall



# Kinematic Filter

Step - 2

## Determining Velocity, Acceleration Threshold

Trained ST-GCN + Validation dataset



**Extract physics features**  
Peak hip velocity and peak acceleration



**Generate candidates from fall data**  
Percentiles of real fall sequences



**Evaluate all candidate pairs**  
Each Fair  $\rightarrow$  Fall F1

Algorithm for determining  
the Thresholds of Vel, Acc & Gray zone

**Select Vel, Acc thresholds**  
Best-scoring pair



**Evaluate gray-zone candidates**  
Each (lower, Upper Threshold : 0.05 - 0.9 )  
pair  $\rightarrow$  Fall F1



**Select gray zone**  
Best scoring pair(lower, upper Threshold)



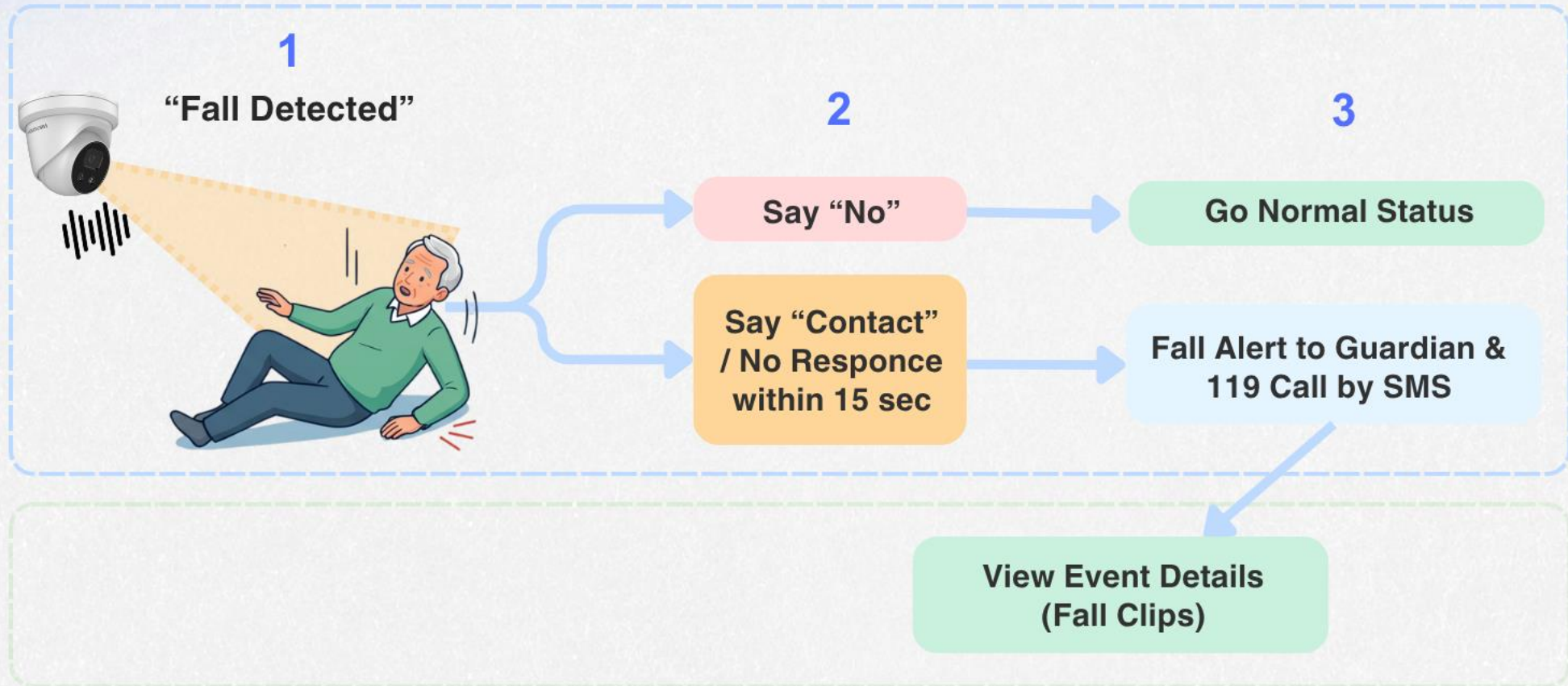
**Final detector configuration**  
All boundaries data-driven

**Determining gray zone**

# TTS & STT

Step - 3

Care Recipient



Guardian



# Dataset



|                   |                                                  |
|-------------------|--------------------------------------------------|
| <b>Subject</b>    | 17                                               |
| <b>Frame</b>      | 589,351                                          |
| <b>Size</b>       | 279 GB                                           |
| <b>Type</b>       | Image                                            |
| <b>Data split</b> | Training : 70%<br>Validation : 15%<br>Test : 15% |

## Challenges in Data Acquisition Systems: Lessons Learned from Fall Detection to Nanosensors

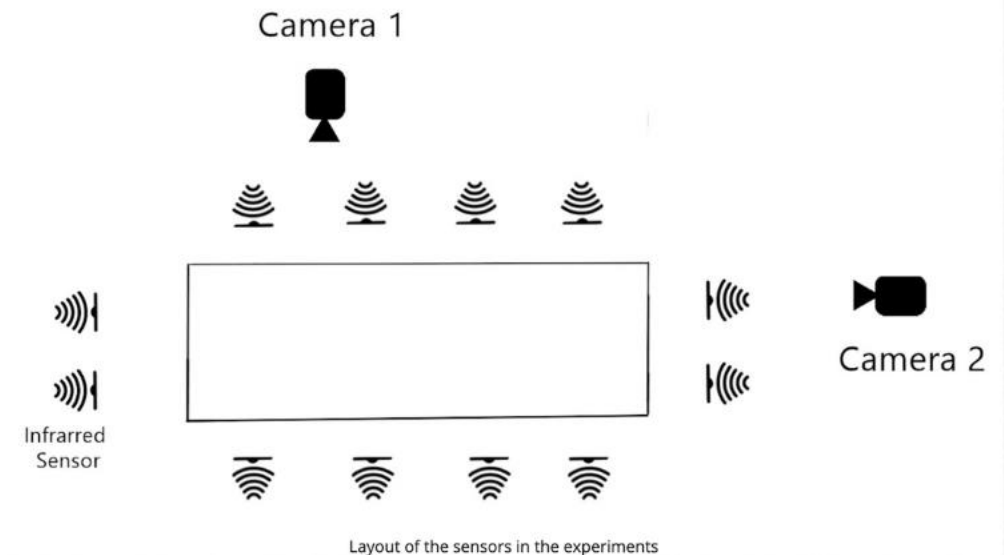
Publisher: IEEE

[Cite This](#)

[PDF](#)

[Carlos J. Peñafort-Asturiano](#) ; [Nestor Santiago](#) ; [José Pablo Núñez-Martínez](#) ; [Hiram Ponce](#) ; [Lourdes Martínez-Villaseñor](#) All Authors

### LAYOUT



# Test Results - Full dataset

## Comparative Table with same dataset

17 Subjects - 19,645 Windows (a Window = 30 Frames)

| Model                               | Accuracy      | F1            | Recall | FP        | FN | FPS   |
|-------------------------------------|---------------|---------------|--------|-----------|----|-------|
| TCN+TE (2025 baseline)<br>(DL + DL) | 0.933         | 0.792         | 0.87   | 137       | 53 | 19    |
| ST-GCN<br>(DL)                      | 0.9873        | 0.9578        | 0.99   | 31        | 5  | 30.48 |
| ST-GCN+ Physics                     | <b>0.9919</b> | <b>0.9723</b> | 0.97   | <b>12</b> | 11 | 30.27 |

If p in Gray zone :  
(0.85 ~ 0.9)



Vel thr : 0.0  
Acc thr : 0.0 } Fall

**Accuracy:** 98.73% → **99.19%**    **F1 Score:** 0.9578 → **0.9723**    **False Positive:** 31 → **12**

When enough training data is available, **ST-GCN already performs well.**

Adding the physics module brings only a small improvement.



What if training data is limited?

# Test Results - Limited dataset

## Comparative Table with same dataset

*3 Subjects - 3,503 Windows (a Window = 30 frames)*

| Model                               | Accuracy      | F1            | Recall       | FP       | FN       | FPS          |
|-------------------------------------|---------------|---------------|--------------|----------|----------|--------------|
| TCN+TE (2025 baseline)<br>(DL + DL) | 0.9230        | 0.789         | 0.973        | 37       | 2        | 19           |
| ST-GCN<br>(DL)                      | 0.9683        | 0.8806        | 0.787        | 0        | 16       | 30.48        |
| <b>ST-GCN+ Physics</b>              | <b>0.9802</b> | <b>0.9315</b> | <b>0.907</b> | <b>3</b> | <b>7</b> | <b>30.27</b> |

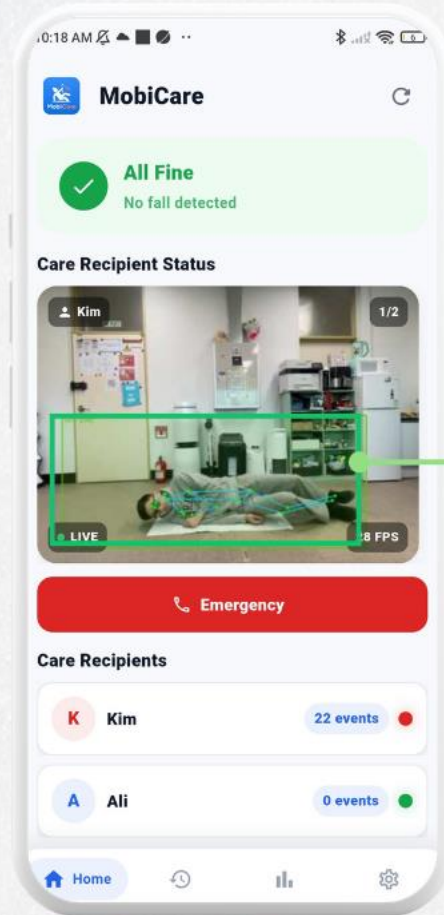
If p in Gray zone :  
(0.1 ~ 0.3)  
↓  
Vel thr : 0.020  
Acc thr : 0.313 } Fall

**Real elderly fall data are difficult to collect.**

The physics module **helps improve** performance when training **data is limited**.

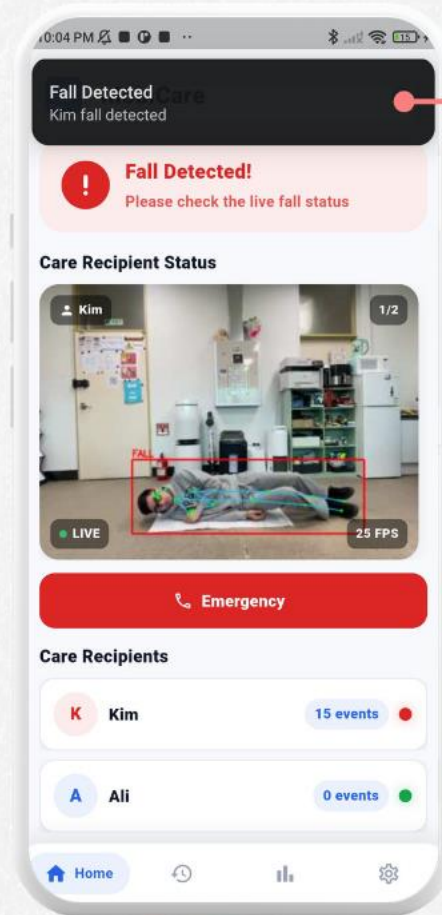


# Mobile App Pages



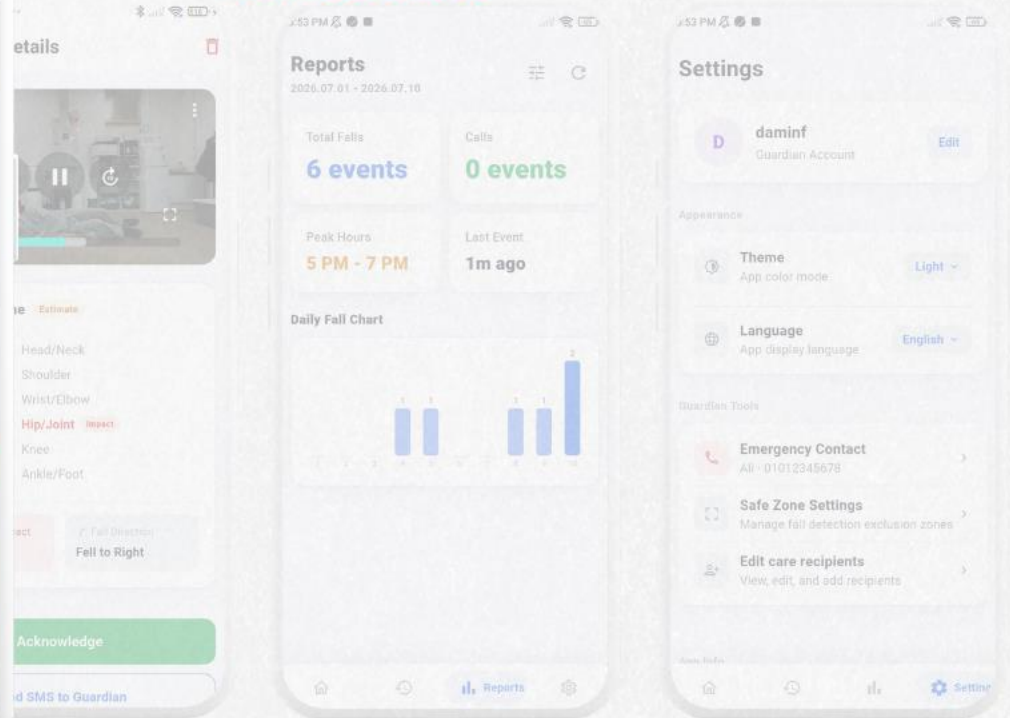
Normal Status

In Safe Zone  
(bed, sofa)  
→ Not Fall



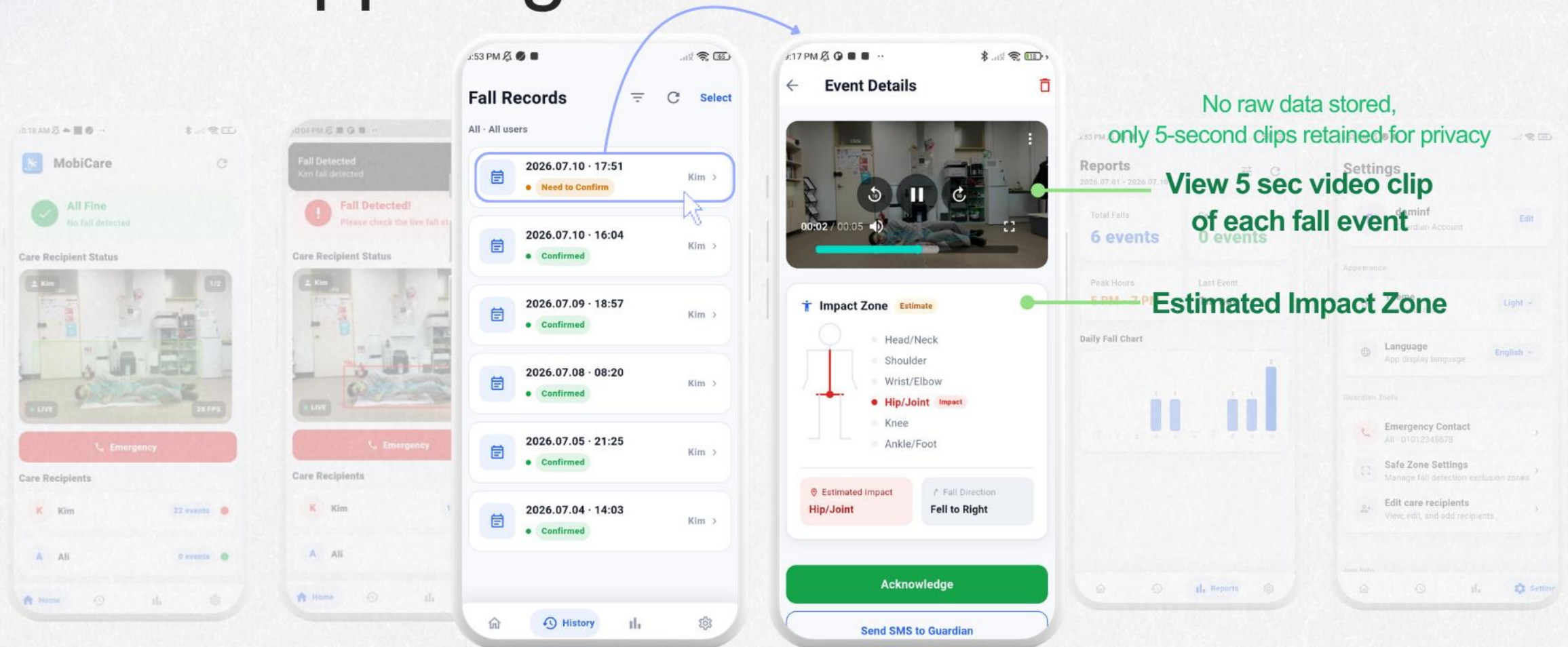
Fall Status

Not in Safe Zone  
→ Fall Detected!



Main Dashboard

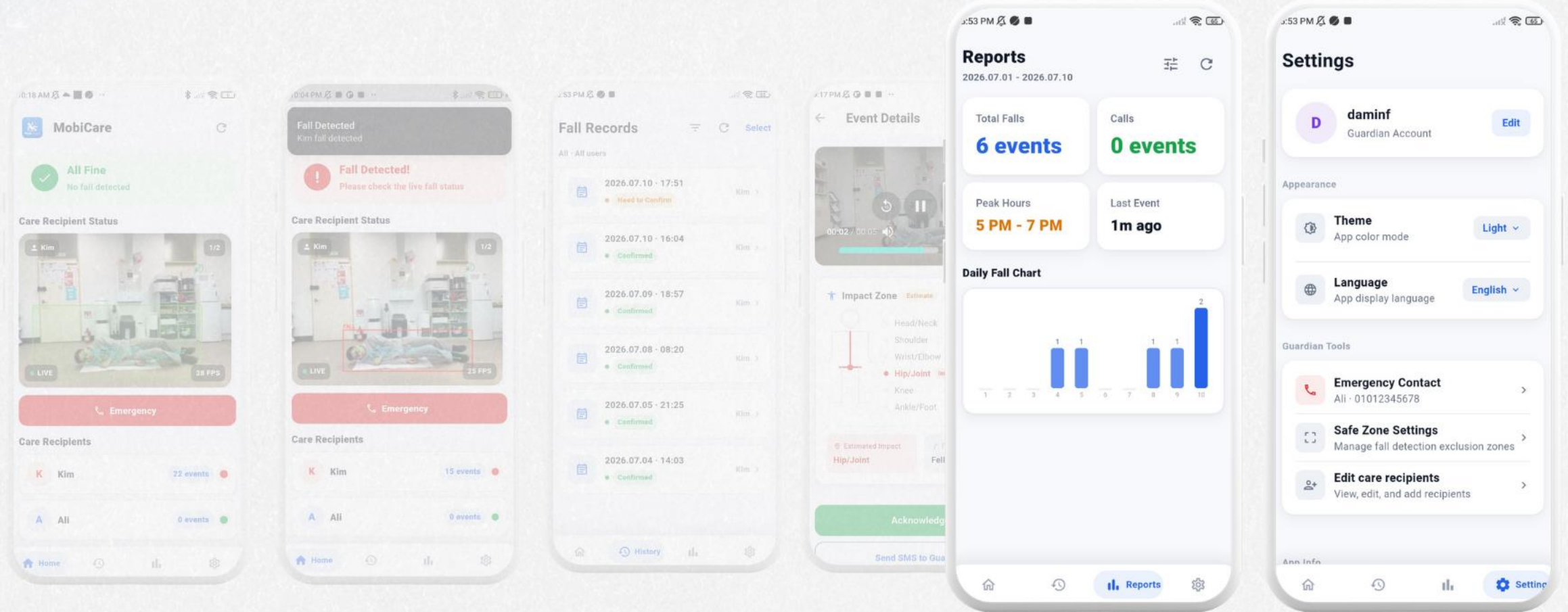
# Mobile App Pages



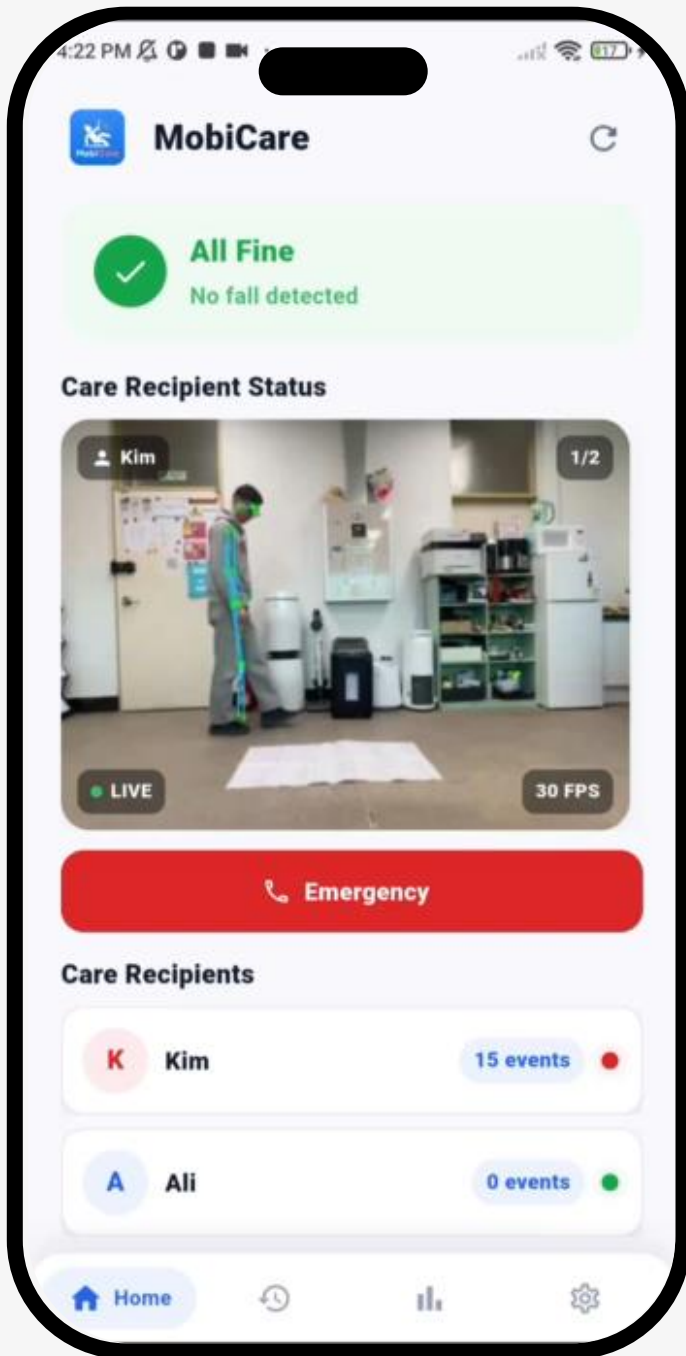
History & Event Details



# Mobile App Pages

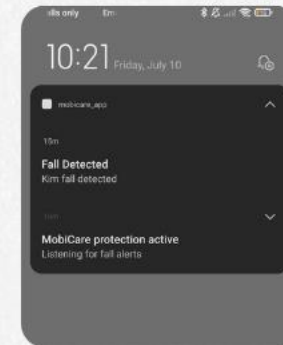
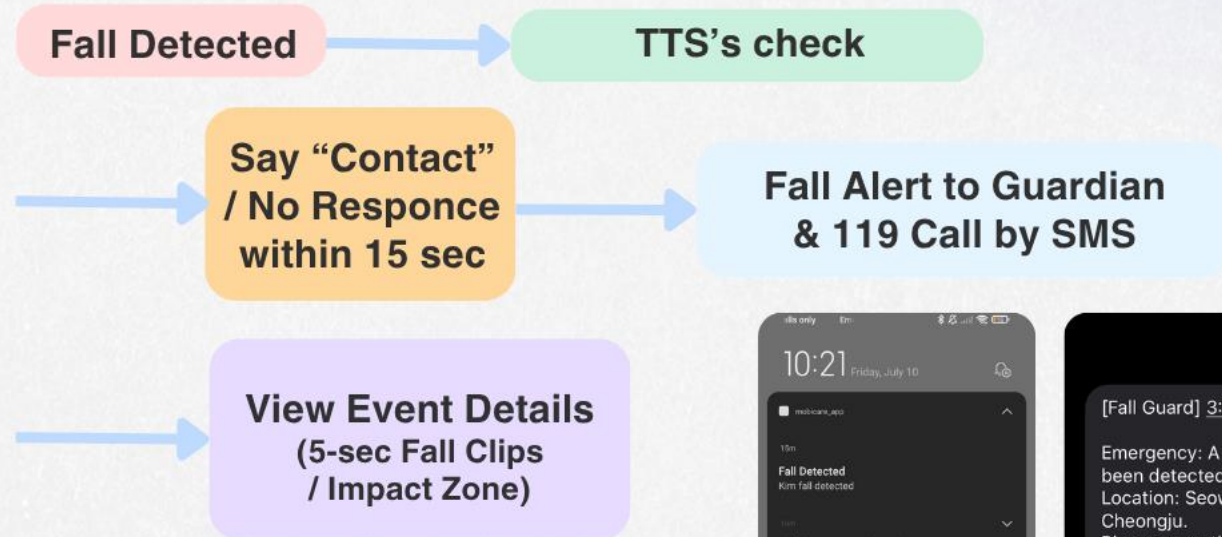


Reports & Settings



# Demo Video

## *(Fall Simulation)*



Push alarm



SMS alarm



# Conclusion and Future Work

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Easily deployable in both institutional and home care settings.

# Thank you!

## Q&A

Team - Choco Haim

Damin Kim, Jonghyeok Park, Aliboev Abbos, Seongwoo Cho

Contact: [ali@chungbuk.ac.kr](mailto:ali@chungbuk.ac.kr)





# Expected Impact and Social Value

## Personal

- Reduced false fall-detection rate
- Detection time: 1 hour to 3 sec
- Community safety support

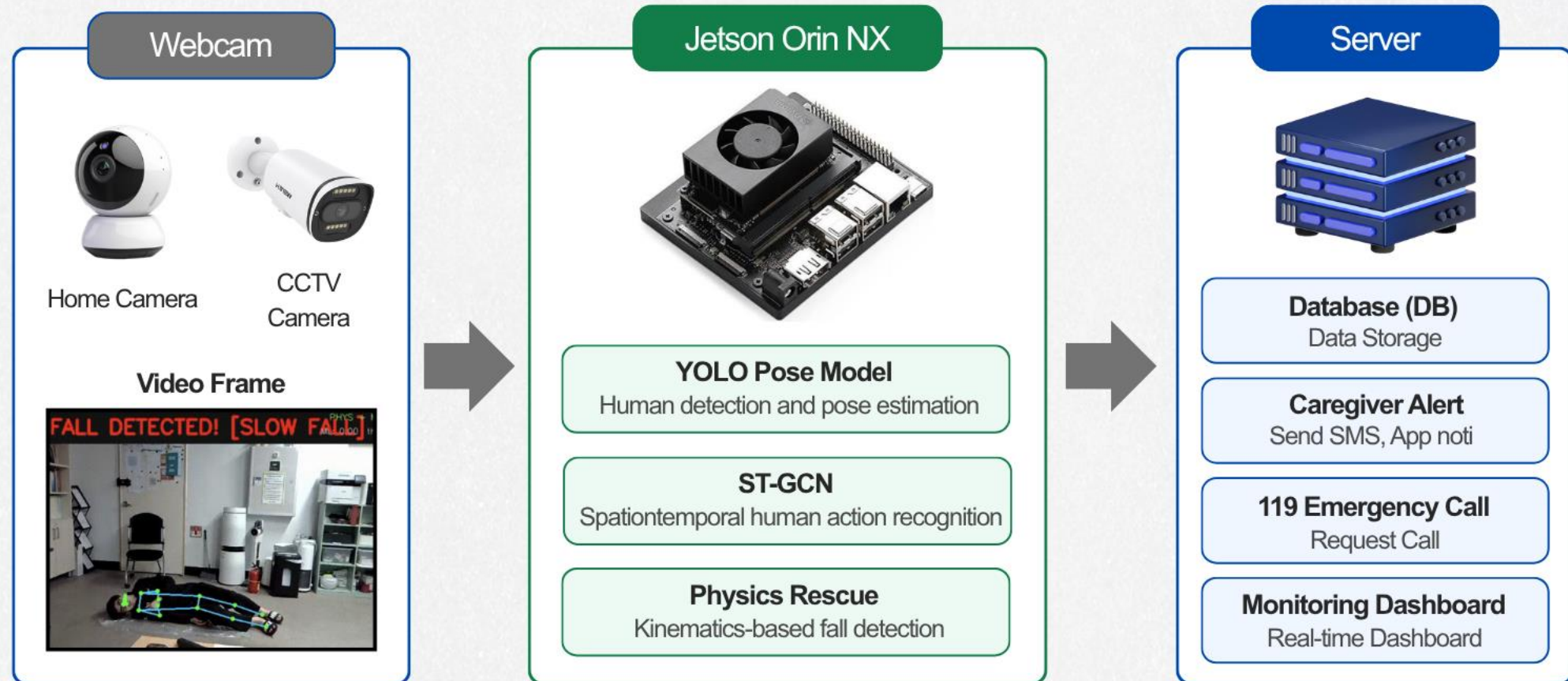
## Social

- Reduced caregiving burden for older adults
- Eased psychological burden on families
- Improved sense of security

## Technical

- Improved detection performance
- Edge-based real-time detection
- Integrated alert system implementation

# System Architecture

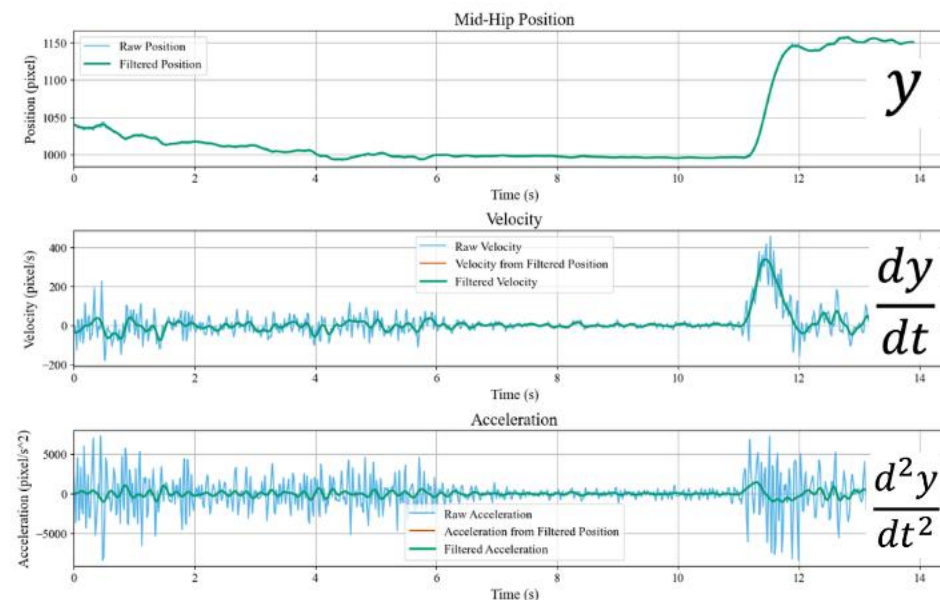
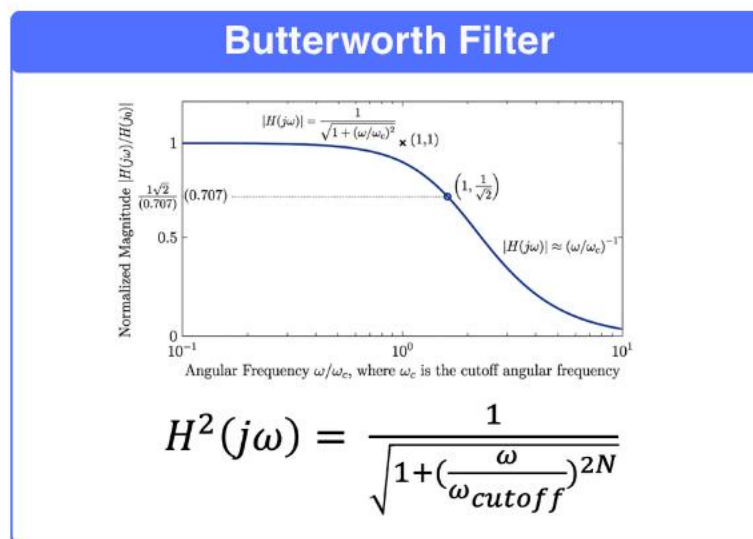




# Physics Rescue

## Step - 2

### Kinematics



- Extraction of mid-hip keypoint pixel coordinates (body's center of mass), with a **fourth-order** Butterworth filter applied for noise removal
- Sequential differentiation to obtain velocity and acceleration, with the Butterworth filter re-applied after each differentiation
- Filter selection rationale: based on the research paper(Michaels et al., 2025)

# Applicability to Elderly Fall Detection

Next, applicability to elderly fall detection. Currently, our model is validated on a limited adult dataset.  
Since the average elderly fall speed is 2.58 meters per second, we can apply this to our model to improve data efficiency.



## Current Status

Validated successfully  
using a **limited adult dataset**.

## Average Elderly Fall Speed

**2.58 m/s**

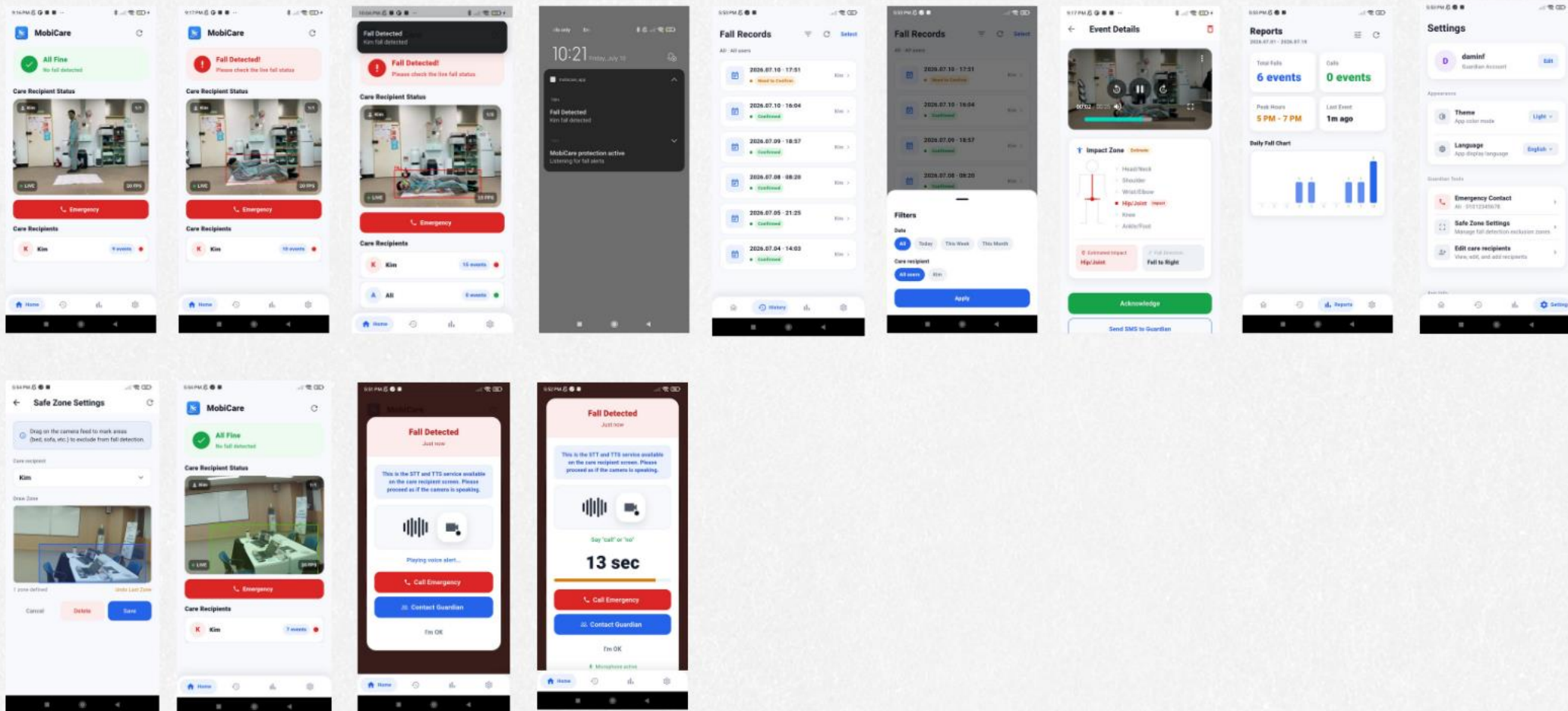
Choi, W. J. (2013). Biomechanics of Falls and Hip Fractures in Older Adults.  
PhD thesis, Simon Fraser University

**Data Efficiency**





# Mobile App



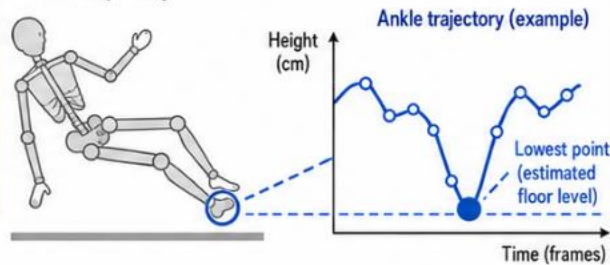
# How Impact Zone Estimated

## How MobiCare's Impact Zone Works

### 1. Estimate Floor Level

For each event, the floor level is estimated from the person's own ankle trajectory.

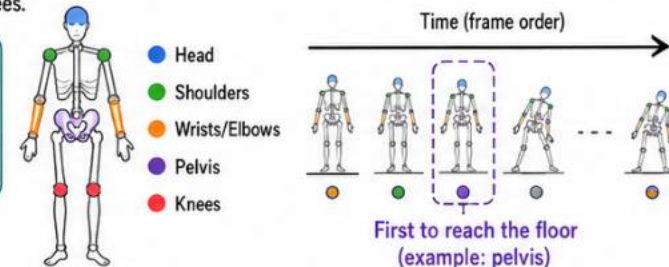
- Use the lowest ankle point during the fall segment
- Independent of camera distance or angle
- No separate calibration required



### 2. Detect Impact Body Part

Check, frame by frame, which body-part group reaches the floor level first among head, shoulders, wrists/elbows, pelvis, and knees.

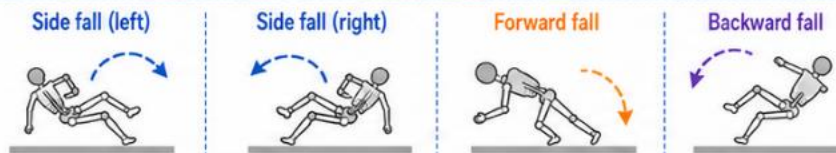
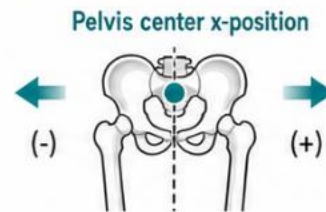
The first part to reach the floor is defined as the impact zone



### 3. Infer Fall Direction

Use pelvis x-position change and impact clues

- |     |                                 |                 |
|-----|---------------------------------|-----------------|
| ←→  | Large left/right pelvis shift   | → Side fall     |
| 👉👈  | Wrist or knee first             | → Forward fall  |
| 👤👤👤 | Head, shoulder, or pelvis first | → Backward fall |



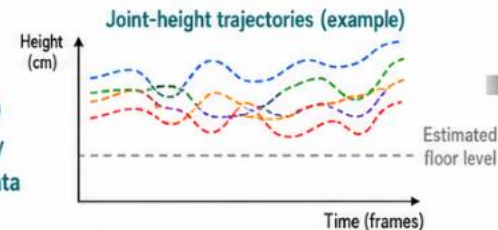
### 4. Honesty Principle

If data are insufficient or no body part gets close to the floor, the function never guesses.

Never guess — return { }



Insufficient / uncertain data



Result

{ }

No guess (inference unavailable)

1 Estimate Floor Level → 2 Detect Impact Body Part → 3 Infer Fall Direction → 4 Honesty Principle